Osptek Display

AMOLED SPECIFICATION

Model No:

AM139Q454454FLS1

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REVISION RECORD

Version	Revision Items	Page	Name	Date
1.0	First release			20250901

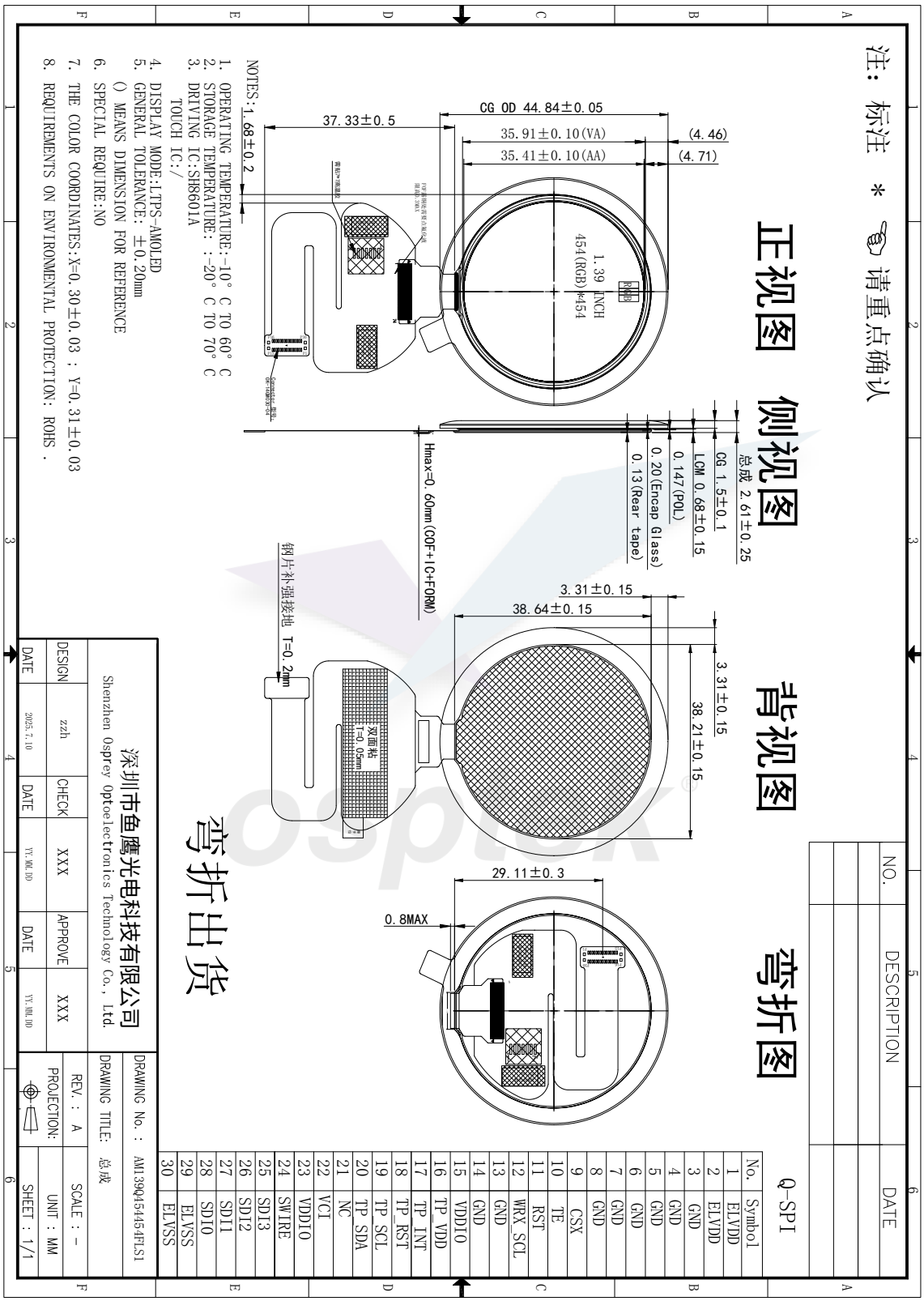
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1.General Spec

Item	Standard values	Unit 单位
Diagonal Inch	1.39	inch
Dot arrangement	454(W) x RGB x 454(H)	dots
Pixel pitch	78 x 78	μ m
Display mode	AMOLED	/
Driver IC	SH8601	/
Module size	44.84(V) x 44.84(W) x 2.61(T)	mm
Active area	φ35.412	mm
Interface	QSPI	-
Operating temperature	-10~60	℃
Storage temperature	-20~70	℃

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2.Mechanical drawing



3. Interface description

Pin No.	Symbol	Description
1-2	ELVDD	AMOLED positive power supply
3-8	GND	Ground.
9	CSX	Chip Select Signal in MIPI DBI Type-B(MPU) and Type-C(SPI) Active Low
10	TE	Tearing effect output pinto synchronize MCU to frame
11	RST	Device reset signal (0 : Enable ; 1: Disable)
12	WRX_SCL	Synchronous Clock Signal in MIPI DBI Type-B(MPU) and Type-C(SPI)
13-14	GND	Ground.
15	VDDIO	Power supply for I/O block provided from outside VDDI < 0.05V (When power is turned off)
16	TPVDD	Touch Panel 3.3Vpower supply
17	TPINT	Touch Panel interrupt
18	TPRST	Touch Panel Reset
19	TPSCL	Touch Panel Clock
20	TPSDA	Touch Panel Data
21	MTP_PWR	NC
22	VCI	Power Supply for Analog Circuits VCI < 0.05V (When power is turned off
23	VDDIO	Power supply for I/O block provided from outside VDDI < 0.05V (When power is turned off)
24	SWIRE	SWIRE signal for PWR IC control
25	SDI3	Serial Data Input in Quad-SPI
26	SDI2	Serial Data Input in Quad-SPI
27	SDI1	Serial Data Input in Quad-SPI
28	SDI0	Serial Data Input in Quad-SPI
29-30	ELVSS	AMOLED negative power supply

4. Absolute Maximum Ratings

Item	Symbol	Min	Max	Unit
Supply voltage (Display)	VDD	-0.3	3.6	V
	IOVCC	-0.3	3.6	V
	ELVDD	4.6±0.05		V
	ELVSS	-2.4±0.05		V
Operating temperature range	Top	-10	+60	℃
Storage temperature range	Tst	-20	+70	℃
Storage humidity	HD	-	90	%RH

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5. DC Characteristics

DC Characteristics for Interface

Table 7 DC Characteristic for Interface Signals

Parameter	Symbol	Condition	Specification			Pin	Unit
			Min.	Typ.	Max.		
Logic high level input voltage	VIH_IO1	–	0.8 × VDDI	–	VDDI	RESX	V
Logic low level input voltage	VIL_IO1	–	0.0	–	0.2 × VDDI		
Logic high level output voltage	VOH_IO1	IOUT = – 1 mA	0.8 × VDDI	–	VDDI	TE	
Logic low level output voltage	VOL_IO1	IOUT = + 1 mA	0.0	–	0.2 × VDDI		
Input high level leakage current	IIH	VIN = VDDI	–	–	1	RESX	μA
Input low level leakage current	IIL	VIN = VSS	– 1	–	–		

NOTE: TA = – 40 to 85 °C

Power Consumption

Table 8 Power Consumption

Parameter	Symbol	Condition	Specification			Unit	Note
			Min.	Typ.	Max.		
Operating current (DSI, dynamic display)	$IVDDI_{op}$	Frame frequency = 60 Hz No load SR_SET = 5'b10110 SPR Function OFF	–	–	2.805	mA	(NOTE)
	$IVCI_{op}$		–	–	5.5		
Sleep current	$IVDDI_{LP}$	LP11 mode	–	–	850	uA	
	$IVCI_{LP}$		–	–	50		
	$IVDDI_{ULPS}$	ULPS mode	–	–	833	uA	
	$IVCI_{ULPS}$		–	–	50		
Deep standby current	$IVDDI_{DSLP}$		–	–	4	uA	
	$IVCI_{DSLP}$		–	–	2		

NOTE: The table above shows only driver IC's power consumption. (MIPI I/F @0.5Gbps, white pattern).
VCI = 2.8V, VDDI = 1.8V, TA = 25 °C.

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Power Characteristic:

Item	Symbol	Min.	Typ.	Max.	Unit	Remark
AMOLED Power positive	ELVDD	4.55	4.6	4.65	V	-
AMOLED power Negative	ELVSS	-2.45	-2.4	-2.35	V	Ref
Digital Power supply	VDDIO	1.65	1.8	3.35	V	Ref
Analog Power supply	VCI	2.7	2.8	3.35	V	Ref

1) Normal Mode

Power Supply: IOVCC=1.8V VCI=2.8V

Frame Frequency: F_{frame} =60HZ @ 25degC, Brightness 350 nits, Command Mode,

Display Condition	Symbol	Min.	Typ.	Max.	Unit	Remark
100% Pixel On 350nits	IELVDD /ELVSS	-	19.3	21.4	mA	Ref
	IVCI	-	6	7.2	mA	Ref
	IVDDIO	-	2	2.4	mA	Ref

2) Idle Mode

Power Supply: IOVCC=1.8V VCI=2.8V

Frame Frequency: F_{frame} =15HZ @ 25degC, Brightness:30nits, OPR:10%,Command Mode

Display Condition	Symbol	Min.	Typ.	Max.	Unit	Remark
10% Pixel On 30 nits	IELVDD /ELVSS	-	-	-	mA	Supplied by Driver IC
	IVCI	-	3	3.6	mA	Ref
	IVDDIO	-	1.2	1.5	mA	Ref

3) Deep Standby Mode

Display Condition	Symbol	Min.	Typ.	Max.	Unit	Remark
Deep Standby	IVCI	-	-	15	uA	-
	IVDDIO	-	-	4	uA	-
	Power Consumption	-	-	<50	uW	Ref

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6.Electron-optical Characteristics

Test condition： IOVCC=1.8V , VCI=2.8V ,Ta=25℃

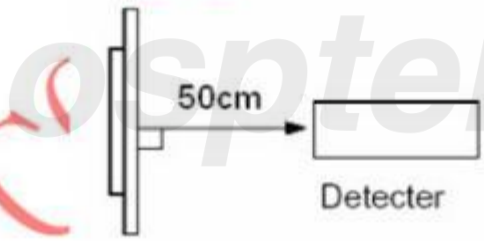
Item		Symbol	Condition	Value			Unit	Note
				Min	Typ	Max		
Luminance			θ=0°	315	350	385	cd/m2	Note 1
Uniformity			Φ=0°	85		-	%	Note 2
Viewing Angle	Left	θ_L	Cr≥200	80	85	-	Deg.	Note 3
	Right	θ_R		80	85	-		
	Top	ψ_T		80	85	-		
	Bottom	ψ_B		80	85	-		
Contrast Ratio		CR	θ=0°	5000	10000	-	-	Note 4

NTSC Ratio	NTSC	CIE1931	97	103	-	%	-
Flicker	-	-	-	-	-30	dB	-
Gamma	-	-	1.9	2.2	2.5		Note 6
Crosstalk	ΔCT	-	-	-	1.1		Note 7

Note 1: Luminance measurement

The test condition is measured on the surface of AMOLED module at 25℃.

- Measurement equipment CS2000 or similar equipment (Field of view:1deg,Distance:50cm)
- Measuring surroundings: Dark room.
- Measuring temperature: Ta=25℃.
- Adjust operating voltage to get optimum contrast at the center of the display.



Note 2: Uniformity

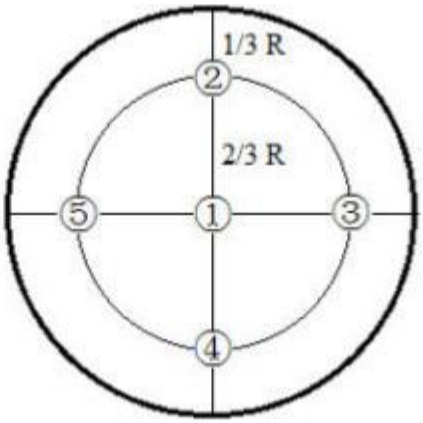
The luminance uniformity is calculated by using following formula:

ΔBp = Bp (Min.) / Bp (Max.)×100 (%)

Bp (Max.) = Maximum brightness in 5 measured spots

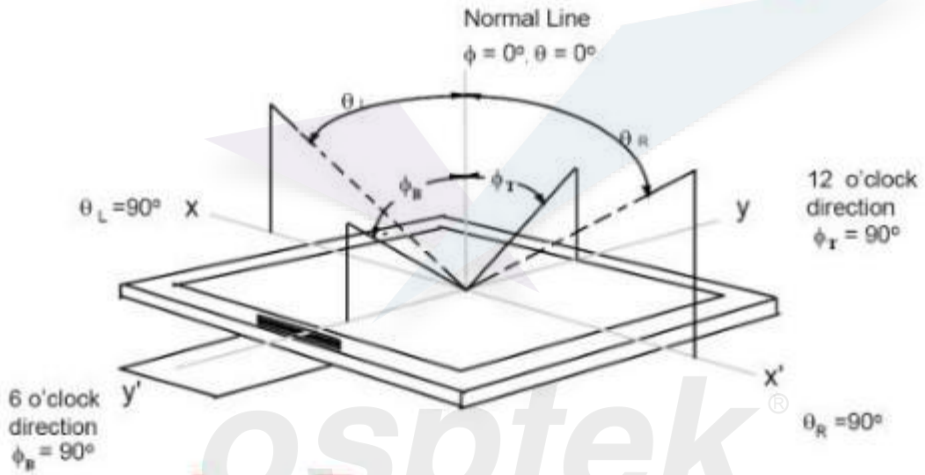
Bp (Min.) = Minimum brightness in 5 measured spots.

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Note 3: **The definition of Viewing Angle**

Refer to the graph below marked by ϑ and Φ



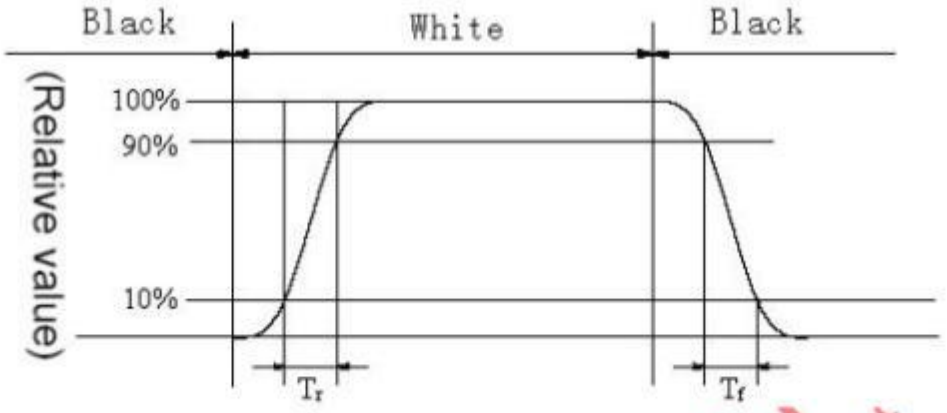
Note 4: **The definition of Contrast Ratio**:

$$\text{Contrast Ratio (CR)} = \frac{\text{Luminance When AMOLED is at "White" state}}{\text{Luminance When AMOLED is at "Black" state}}$$

Note 5: **Definition of Response time.**

The output signals of photo detector are measured when the input signals are changed from "black" to "white" (Voltage falling time) and from "white" to "black" (Voltage rising time), respectively. The response time is defined as the time interval between the 10% and 90% of amplitudes. Refer to figure as below.

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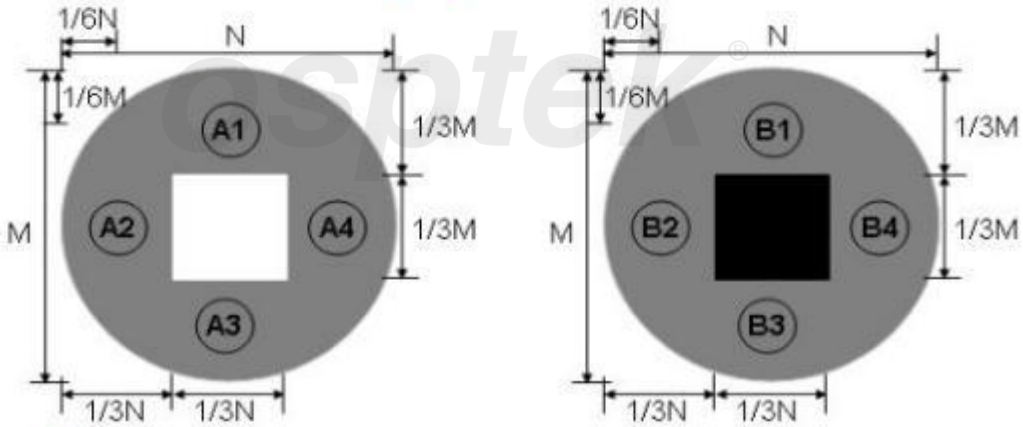
Note 6: Gamma curve

The whole curve's tolerance must control within +/-0.3, test the gray scale below:
8, 16, 25, 33, 41, 49, 58, 66, 74, 82, 90, 99, 107, 115, 123, 132, 140, 148, 156, 165, 173, 181, 189, 197,206, 214, 222, 230, 239, 255

Note 7: **Crosstalk**

There should be no visible cross-talk in normal direction of the display when the two "Cross-talk Test Patterns" below are loaded.

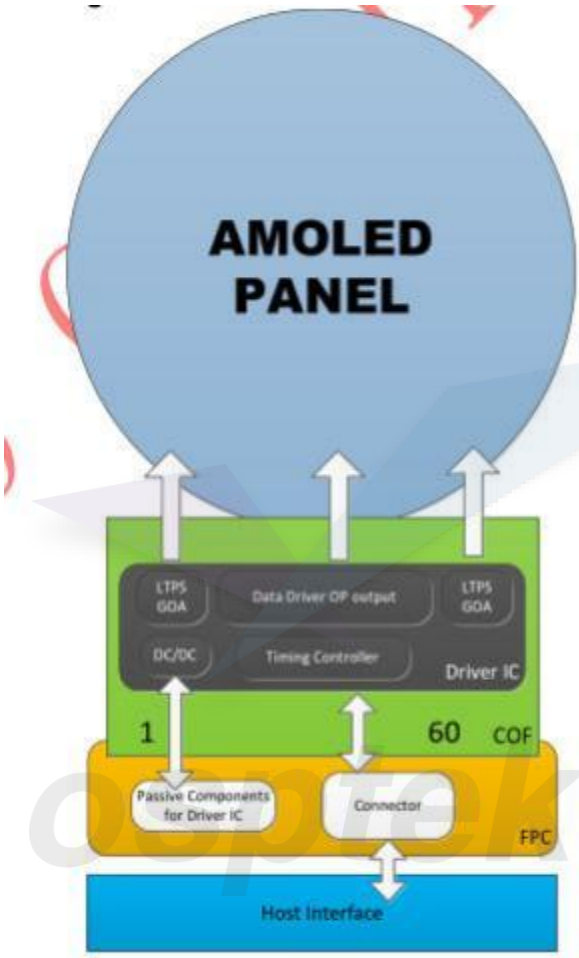
ΔBp (Max.) = Maximum value in ΔBp1~ΔBp4.
ΔBp (Min.) = Minimum value in ΔBp1~ΔBp4.
ΔCT=ΔBp (Max.)/ ΔBp(Min.).
ΔCT must be less than 1.10



Cross-talk Test Pattern

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7. System Block Diagram



8. AC characteristic& Power On/Off Sequence

Power On/Off Sequence

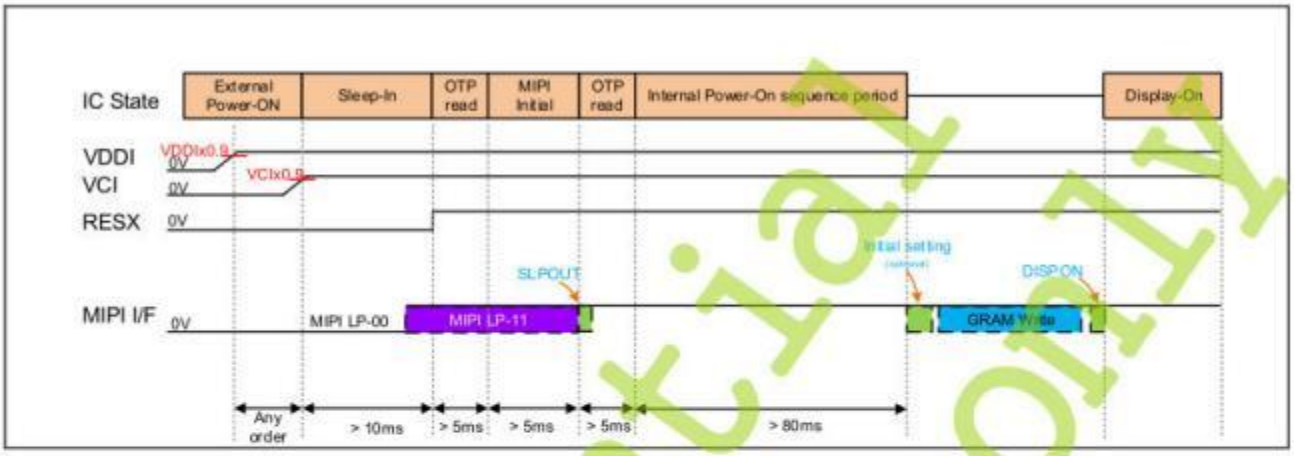


Figure 16 Power-On Sequence



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3.1.1 Quad SPI

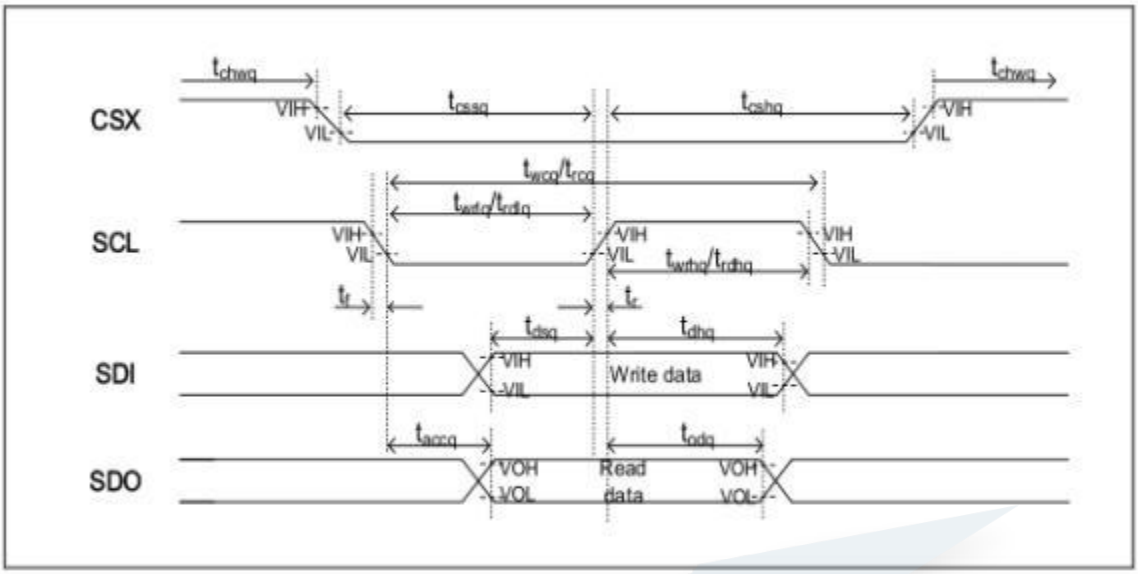


Figure 6 Quad Serial Interface Characteristics

Table 3 AC Characteristics of Quad SPI

Characteristic	Symbol		Specification		Unit
			Min.	Max.	
Chip select setup time	CSX	t_{cssq}	40	–	ns
Chip select hold time		t_{cshq}	40	–	ns
Chip select "High" pulse width		t_{chwq}	50	–	ns
Write cycle time	SCL (Write)	t_{wcq}	20	–	ns
SCL "High" period (Write)		t_{wrhq}	25	–	ns
SCL "Low" period (Write)		t_{wrlq}	25	–	ns
Read cycle time	SCL (Read)	t_{rcq}	100	–	ns
SCL "High" period (Read)		t_{rdhq}	50	–	ns
SCL "Low" period (Read)		t_{rdlq}	50	–	ns
Data setup time	SDI	t_{dsq}	15	–	ns
Data hold time		t_{dhq}	15	–	ns
Access time	SDO	t_{accq}	5	35	ns
Output disable time		t_{odq}	20	–	ns
Rise/Fall time	-	t_r/t_f	–	1	ns

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9. Quad SPI Interface&Quad SPI Interface Protocol – Register Read and Write

Table 29 Quad SPI PAD Assignment

PAD name	IO	DBI Type-B (80-series 8bit MPU)		
		Signal Name	IO	Description
CSX	I	CSX	I	Chip selection, Active Low
SCL_WRX	I	SCL	I	Serial clock, Rising edge
SDI_RDX	IO	SDIO	IO	Serial Data in 0, Data out
SDO_SDI1	IO	SDI1	I	Serial Data input 1
SDI2	IO	SDI2	I	Serial Data input 2
SDI3	IO	SDI3	I	Serial Data input 3

Quad SPI Interface

In addition to MIPI DBI Type-C interface, SH8601A supports Quad-SPI interface. Register read, write interface and RGB pixel interface timing diagrams are shown below. 1-wire or 4-wire option is selected by register SPI_MODE(C4h).

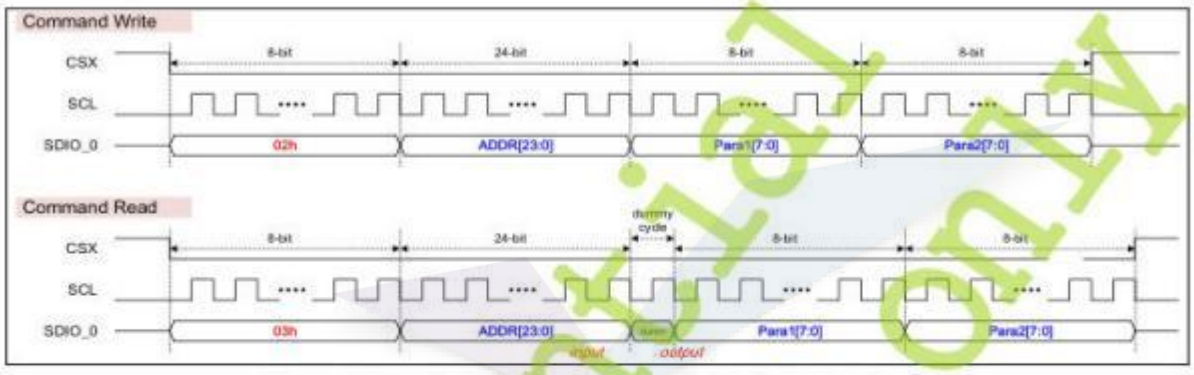


Figure 34 Quad SPI Interface Protocol – Register Read and Write



Figure 35 Quad SPI Interface Protocol – Pixel Interface

10.Reliability test conditions and methods

No	Test Item	Test condition	Criterion
1	High Temperature Storage	70℃±3℃ 48H Power off	Inspection after 4 hours storage at room temperature,the sample shall be free from defects: 1. Air bubble in the LCD; 2.Seal leak; 3.Non-display; 4.missing segments; 5.Glass crack; 6.Current Idd is twice higher than initial value. 7.The surface shall be free from damage. 8.The electric characteristic requirements shall be satisfied
2	Low Temperature Storage	-20℃±3℃ 48H Power off	
3	High Temperature Operation	60℃±3℃ 48H Power on	
4	Low Temperature Operation	-10℃±3℃ 48H Power on	
5	High Temperature & Humidity Operation	60℃±3℃ 90%RH 48H Power off	
6	Temperature Cycle	-20℃←→25℃←→70℃ 30min 5min 30min 5 cycle Power off	

Remark:

1.The test samples should be applied to only one test item.

2.Sample size for each test item is 3~5pcs.

3.For Damp Proof Test, Pure water(Resistance > 10MΩ) should be used.

4.In case of malfunction defect caused by ESD damage, if it would be recovered to normal state after resetting, it would be judged as a good part.

5.EL evaluation should be excepted from reliability test with humidity and temperature: Some defects such as black spot/blemish can happen

by natural chemical reaction with humidity and Fluorescence EL has.

6.Failure Judgment Criterion: Basic Specification, Electrical Characteristic, Mechanical Characteristic, Optical Characteristic.